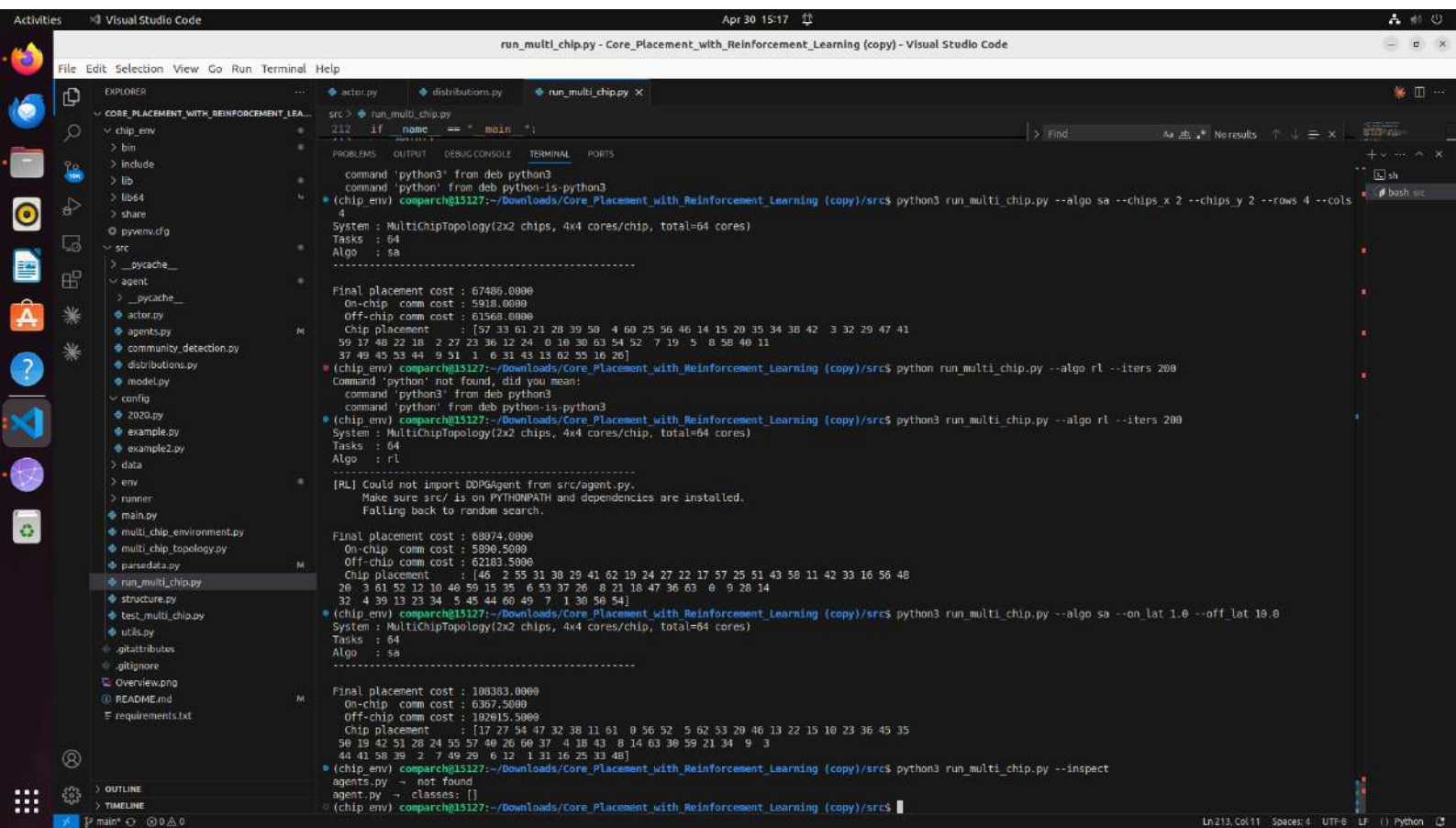




Visual Studio Code interface showing a Python project named "run_multi_chip.py - Core Placement with Reinforcement Learning (copy)". The Explorer pane on the left shows the project structure, including files like `actor.py`, `distributions.py`, `run_multi_chip.py`, `__pycache__`, `agent`, `agents.py`, `community_detection.py`, `distributions.py`, `model.py`, `config`, `2020.py`, `example.py`, `example2.py`, `data`, `env`, `runner`, `main.py`, `multi_chip_environment.py`, `multi_chip_topology.py`, `parsedata.py`, `run_multi_chip.py`, `structure.py`, `test_multi_chip.py`, `utils.py`, `.gitattributes`, `.gitignore`, `Overview.png`, `README.md`, and `requirements.txt`.

The main editor displays the `run_multi_chip.py` file, which contains a `main()` function. The code defines a `MultiChipTopology` object and runs a reinforcement learning algorithm. The output shows the results of the algorithm, including the final placement cost, on-chip communication cost, off-chip communication cost, and chip placement data.

```
def main():
    (chip.env) comparch@15127:~/Downloads/Core Placement with Reinforcement Learning (copy)/src$ python3 run_multi_chip.py --algo ppo
System : MultiChipTopology(2x2 chips, 4x4 cores/chip, total=64 cores)
Tasks : 64
Algo : ppo

[PP0] Could not import PPOAgent: No module named 'agents'
Falling back to SA.
SA step 500 | cost=66383.00 | on=6331.50 off=60299.50
SA step 1000 | cost=66161.00 | on=6341.00 off=60808.50
SA step 1500 | cost=66161.00 | on=6273.50 off=60170.50
SA step 2000 | cost=66161.00 | on=6259.00 off=60254.50
SA step 2500 | cost=66161.00 | on=6276.50 off=60131.50
SA step 3000 | cost=66161.00 | on=6296.00 off=60102.00
SA step 3500 | cost=65349.50 | on=6408.50 off=59117.00
SA step 4000 | cost=64713.00 | on=6265.50 off=59129.50
SA step 4500 | cost=64549.50 | on=6376.50 off=58485.00
SA step 5000 | cost=64399.50 | on=6397.00 off=58101.50

Final placement cost : 64399.5000
On-chip comm cost : 6486.5000
Off-chip comm cost : 57982.0000
Chip placement : [ 8 34 18 27 44 50 43 31 46 58 63 39 3 24 15 49 0 13 7 38 9 57 62 28
21 33 19 29 47 12 20 16 59 5 60 51 46 14 55 56 42 4 11 26 10 61 6 35
48 52 22 54 23 53 17 1 41 32 36 2 45 25 30 37]

(chip.env) comparch@15127:~/Downloads/Core Placement with Reinforcement Learning (copy)/src$ python3 run_multi_chip.py --algo ppo --chips x 2 --chips y 2 --rows 4 --col
s 4 --off lot 10.0
System : MultiChipTopology(2x2 chips, 4x4 cores/chip, total=64 cores)
Tasks : 64
Algo : ppo

[PP0] Could not import PPOAgent: No module named 'agents'
Falling back to SA.
SA step 500 | cost=111375.50 | on=6202.00 off=105849.50
SA step 1000 | cost=109897.50 | on=6205.00 off=103765.00
SA step 1500 | cost=108513.50 | on=6411.00 off=102102.50
SA step 2000 | cost=107947.50 | on=6395.00 off=102711.50
SA step 2500 | cost=107689.00 | on=6544.00 off=101611.00
SA step 3000 | cost=107247.00 | on=6455.00 off=101449.00
SA step 3500 | cost=107197.50 | on=6411.50 off=101746.00
SA step 4000 | cost=107106.50 | on=6486.00 off=100947.50
SA step 4500 | cost=106986.50 | on=6381.00 off=101432.00
SA step 5000 | cost=106981.50 | on=6445.00 off=101205.50

Final placement cost : 106981.5000
On-chip comm cost : 6592.0000
Off-chip comm cost : 100479.5000
Chip placement : [10 58 25 28 44 54 47 16 40 57 59 36 29 17 0 62 6 7 3 38 13 60 59 19
26 55 52 11 35 8 27 21 32 9 56 53 15 2 51 46 33 30 12 23 4 43 5 45
63 46 18 61 49 39 31 1 42 37 34 29 14 22 24 41]
```

Visual Studio Code interface showing a Python project for Core Placement with Reinforcement Learning.

File Explorer: The project structure includes a `chip_env` directory with subdirectories `bin`, `include`, `lib`, `lib64`, and `share`. The `src` directory contains `__pycache__`, `agent`, `actor.py`, `agents.py`, `community_detection.py`, `distributions.py`, `model.py`, `config`, `2020.py`, `example.py`, `example2.py`, `data`, `env`, `runner`, `main.py`, `multi_chip_environment.py`, `multi_chip_topology.py`, `parsedata.py`, `run_multi_chip.py`, `structure.py`, `test_multi_chip.py`, `utils.py`, `.gitattributes`, `.gitignore`, `Overview.png`, `README.md`, and `requirements.txt`.

Code Editor: The file `run_multi_chip.py` is open. It defines an entry point for multi-chip core placement, listing algorithms: `ppo` (uses existing PPOAgent), `sa` (Simulated Annealing), and `random` (Random search baseline). A quick start section shows how to run the script with various flags.

```
1 """
2 run_multi_chip.py
3 -----
4 Entry point for multi-chip core placement.
5
6 Algorithms:
7   ppo : Uses the existing PPOAgent from agents.py (fully integrated)
8   sa  : Simulated Annealing (no ML deps needed)
9   random : Random search baseline
10
11 Quick start:
12   python3 run_multi_chip.py --algo sa
13   python3 run_multi_chip.py --algo ppo
14   python3 run_multi_chip.py --algo ppo --chips_x 2 --chips_y 2 --rows 4 --cols 4
15 """
16
17 import argparse
18 import math
19 import random
20 import sys
```

Terminal: The terminal shows the execution of the script. It reports the system configuration (MultiChipTopology(2x2 chips, 4x4 cores/chip, total=64 cores)), the number of tasks (64), and the algorithm used (ppo). The output shows the cost and off-chip cost for each step, and the final placement cost.

```
(chip_env) comparch@15127:~/Downloads/Core_Placement_with_Reinforcement_Learning (copy)/src$ python3 run_multi_chip.py --algo ppo --chips_x 2 --chips_y 2 --rows 4 --cols 4 --off lat 10.0
System : MultiChipTopology(2x2 chips, 4x4 cores/chip, total=64 cores)
Tasks : 64
Algo : ppo
-----
[PP0] Could not import PPOAgent: No module named 'agents'
Falling back to SA.
SA step 500 | cost=111375.50 | on=6202.00 off=105049.50
SA step 1000 | cost=109897.50 | on=6205.00 off=103765.00
SA step 1500 | cost=108513.50 | on=6411.00 off=102102.50
SA step 2000 | cost=107947.50 | on=6295.00 off=102711.50
SA step 2500 | cost=107609.00 | on=6544.00 off=101611.00
SA step 3000 | cost=107247.00 | on=6455.00 off=101449.00
SA step 3500 | cost=107197.50 | on=6411.50 off=101746.00
SA step 4000 | cost=107106.50 | on=6486.00 off=100947.50
SA step 4500 | cost=106986.50 | on=6381.00 off=101432.00
SA step 5000 | cost=106981.50 | on=6445.00 off=101205.50

Final placement cost : 106981.5000
On-chip comm cost : 6502.0000
Off-chip comm cost : 100479.5000
Chip placement : [10 58 25 28 44 54 47 16 40 57 50 36 20 17 0 62 6 7 3 38 13 60 59 19
26 55 52 11 35 8 27 21 32 9 56 53 15 2 51 46 33 30 12 23 4 43 5 45
63 46 18 61 49 39 31 1 42 37 34 29 14 22 24 41]
(chip_env) comparch@15127:~/Downloads/Core_Placement_with_Reinforcement_Learning (copy)/src$ python3 run_multi_chip.py --algo ppo
```

Visual Studio Code interface showing a Python project for reinforcement learning.

File Explorer:

- core_placement_with_reinforcement_learning
 - bin
 - include
 - lib
 - lib64
 - share
 - pyvenv.cfg
 - src
 - __pycache__
 - agent
 - __pycache__
 - actor.py
 - agents.py
 - community_detection.py
 - distributions.py
 - model.py
 - config
 - 2020.py
 - example.py
 - example2.py
 - data
 - env
 - runner
 - main.py
 - multi_chip_environment.py
 - multi_chip_topology.py
 - parsedata.py
 - run_multi_chip.py
 - structure.py
 - test_multi_chip.py
 - utils.py
 - .gitattributes
 - .gitignore
 - Overview.png
 - README.md
 - requirements.txt

Code Editor: agents.py

```
11 class PPOAgent:
12     def __init__(self, env, model, device):
13         self.env = env
14         self.model = model
15         self.device = device
16         self.reset_buffer()
17         self.log_probs.append(
18             self.model.log_prob(self.env.reset())
19         )
20         self.rewards.append(reward)
21
22     def train(self) -> None:
23         """Train agent."""
24         best_reward = 1e12
25
26         epoch = 1
27         while epoch <= 100 and best_reward > 768 :
28             self.reset_buffer()
29             for _ in range(self.batch_size):
30                 # reward sampling
31                 with torch.no_grad():
32                     action, log_prob = self.model(self.env.reset())
33                     reward, current_grid = self.env.step(action)
34
35                 # store episodes
36                 self.save_episode(action, log_prob, reward)
37                 if (reward < best_reward):
38                     best_reward = reward
```

Terminal:

```
# of epochs: 482, reward mean: 69117.33 (70605.00)
# of epochs: 483, reward mean: 69136.84 (70605.00)
# of epochs: 484, reward mean: 69116.09 (70605.00)
# of epochs: 485, reward mean: 69193.04 (70605.00)
# of epochs: 486, reward mean: 69978.32 (70605.00)
# of epochs: 487, reward mean: 69148.16 (70605.00)
# of epochs: 488, reward mean: 69182.69 (70605.00)
# of epochs: 489, reward mean: 69985.39 (70605.00)
# of epochs: 490, reward mean: 69946.78 (70605.00)
# of epochs: 491, reward mean: 69978.86 (70605.00)
# of epochs: 492, reward mean: 69990.26 (70605.00)
# of epochs: 493, reward mean: 69992.96 (70605.00)
# of epochs: 494, reward mean: 69176.91 (70605.00)
# of epochs: 495, reward mean: 69952.34 (70605.00)
# of epochs: 496, reward mean: 69101.36 (70605.00)
# of epochs: 497, reward mean: 69176.09 (70605.00)
# of epochs: 498, reward mean: 69126.52 (70605.00)
# of epochs: 499, reward mean: 69120.41 (70605.00)
# of epochs: 500, reward mean: 69968.95 (70605.00)

Final placement cost : 70605.0000
On-chip comm cost : 5453.0000
Off-chip comm cost : 65152.0000
Chip placement : [49, 1 24 25 33, 9 36 44 18 41 31 46 15 38, 3 59 34 63 14 47 50 40, 0 10
54 56 30 35, 8 57 28 27 26 17 52 21, 2 61 12 19 32 43 11 39 45, 4 55 53
62, 7 48 37 42 29 58, 5 51 22 13 20 60 16, 6 23]
```

Visual Studio Code interface showing a project named "agents.py - Core Placement with Reinforcement Learning (copy)". The Explorer sidebar on the left displays the project structure, including folders like "chip_env", "bin", "include", "lib", "share", "pyvenv.cfg", "src", "config", "data", "env", "runner", and files like "main.py", "multi_chip_environment.py", "multi_chip_topology.py", "parsedata.py", "run_multi_chip.py", "structure.py", "test_multi_chip.py", "utils.py", ".gitattributes", ".gitignore", "Overview.png", "README.md", and "requirements.txt".

The main editor displays the `agents.py` file, which defines the `PP0Agent` class. The code includes a `train` method that iterates over epochs, resets the buffer, and performs training steps. The `train` method is as follows:

```
11 class PP0Agent:
12     def __init__(self):
13         self.log_probs.append(
14             self.reset_buffer()
15         )
16         self.rewards.append(reward)
17
18     def train(self) -> None:
19         """Train agent."""
20         best_reward = 1e12
21
22         epoch = 1
23         while epoch <= 100 and best_reward > 768 :
24             self.reset_buffer()
25             for _ in range(self.batch_size):
26                 # reward sampling
27                 with torch.no_grad():
28                     action, log_prob = self.policy()
29                     reward, current_grid = self.env.step(action)
30
31                 # store episodes
32                 self.save_episode(action, log_prob, reward)
33                 if(reward < best_reward):
```

The bottom panel shows the OUTPUT window, displaying training progress and final placement costs:

```
# of epochs: 482, reward mean: 69117.33 (70605.00)
# of epochs: 483, reward mean: 69136.84 (70605.00)
# of epochs: 484, reward mean: 69116.09 (70605.00)
# of epochs: 485, reward mean: 69193.04 (70605.00)
# of epochs: 486, reward mean: 69978.32 (70605.00)
# of epochs: 487, reward mean: 69148.16 (70605.00)
# of epochs: 488, reward mean: 69182.69 (70605.00)
# of epochs: 489, reward mean: 69985.39 (70605.00)
# of epochs: 490, reward mean: 69946.78 (70605.00)
# of epochs: 491, reward mean: 69978.86 (70605.00)
# of epochs: 492, reward mean: 69990.26 (70605.00)
# of epochs: 493, reward mean: 69992.96 (70605.00)
# of epochs: 494, reward mean: 69176.91 (70605.00)
# of epochs: 495, reward mean: 69952.34 (70605.00)
# of epochs: 496, reward mean: 69101.36 (70605.00)
# of epochs: 497, reward mean: 69176.09 (70605.00)
# of epochs: 498, reward mean: 69126.52 (70605.00)
# of epochs: 499, reward mean: 69120.41 (70605.00)
# of epochs: 500, reward mean: 69968.95 (70605.00)

Final placement cost : 70605.0000
On-chip comm cost : 5453.0000
Off-chip comm cost : 65152.0000
Chip placement : [49, 1 24 25 33, 9 36 44 18 41 31 46 15 38, 3 59 34 63 14 47 50 40, 0 10
54 56 30 35, 8 57 28 27 26 17 52 21, 2 61 12 19 32 43 11 39 45, 4 55 53
62, 7 48 37 42 29 58, 5 51 22 13 20 60 16, 6 23]
```

The status bar at the bottom indicates the file is at line 62, column 27, with 4 spaces, UTF-8 encoding, LF line endings, and a Python interpreter.

Visual Studio Code interface showing a project named "agents.py - Core Placement with Reinforcement Learning (copy)". The Explorer sidebar on the left lists the project structure, including folders like "chip_env", "src", and "config", and files like "agents.py", "compare_algos.py", and "README.md".

The main editor displays the code for `agents.py`, which defines a `PP0Agent` class. The class includes methods for saving episodes, training, and a `train` method that runs for 100 epochs. The code is as follows:

```
11 class PP0Agent:
12     def save_episode(
13         log_prob: tuple[List[int], List[int]],
14         reward: float
15     ) -> None:
16         """Store episode in buffer."""
17         self.actions.append(
18             (action[0].cpu().tolist(), action[1].cpu().tolist())
19         )
20         self.log_probs.append(
21             (log_prob[0].cpu().tolist(), log_prob[1].cpu().tolist())
22         )
23         self.rewards.append(reward)
24
25     def train(self) -> None:
26         """Train agent."""
27         best_reward = 1e12
28
29         epoch = 1
30         while epoch <= 100 and best_reward > 768 :
31             self.reset_buffer()
32             for _ in range(self.batch_size):
```

The bottom panel shows the output of the training process, displaying the reward mean for each epoch from 491 to 500. The output is as follows:

```
# of epochs: 491, reward mean: 69966.32 (70461.50)
# of epochs: 492, reward mean: 69120.55 (70461.50)
# of epochs: 493, reward mean: 69173.94 (70461.50)
# of epochs: 494, reward mean: 69830.91 (70461.50)
# of epochs: 495, reward mean: 69121.24 (70461.50)
# of epochs: 496, reward mean: 69141.05 (70461.50)
# of epochs: 497, reward mean: 69967.05 (70461.50)
# of epochs: 498, reward mean: 69157.12 (70461.50)
# of epochs: 499, reward mean: 69103.23 (70461.50)
# of epochs: 500, reward mean: 69115.96 (70461.50)
```

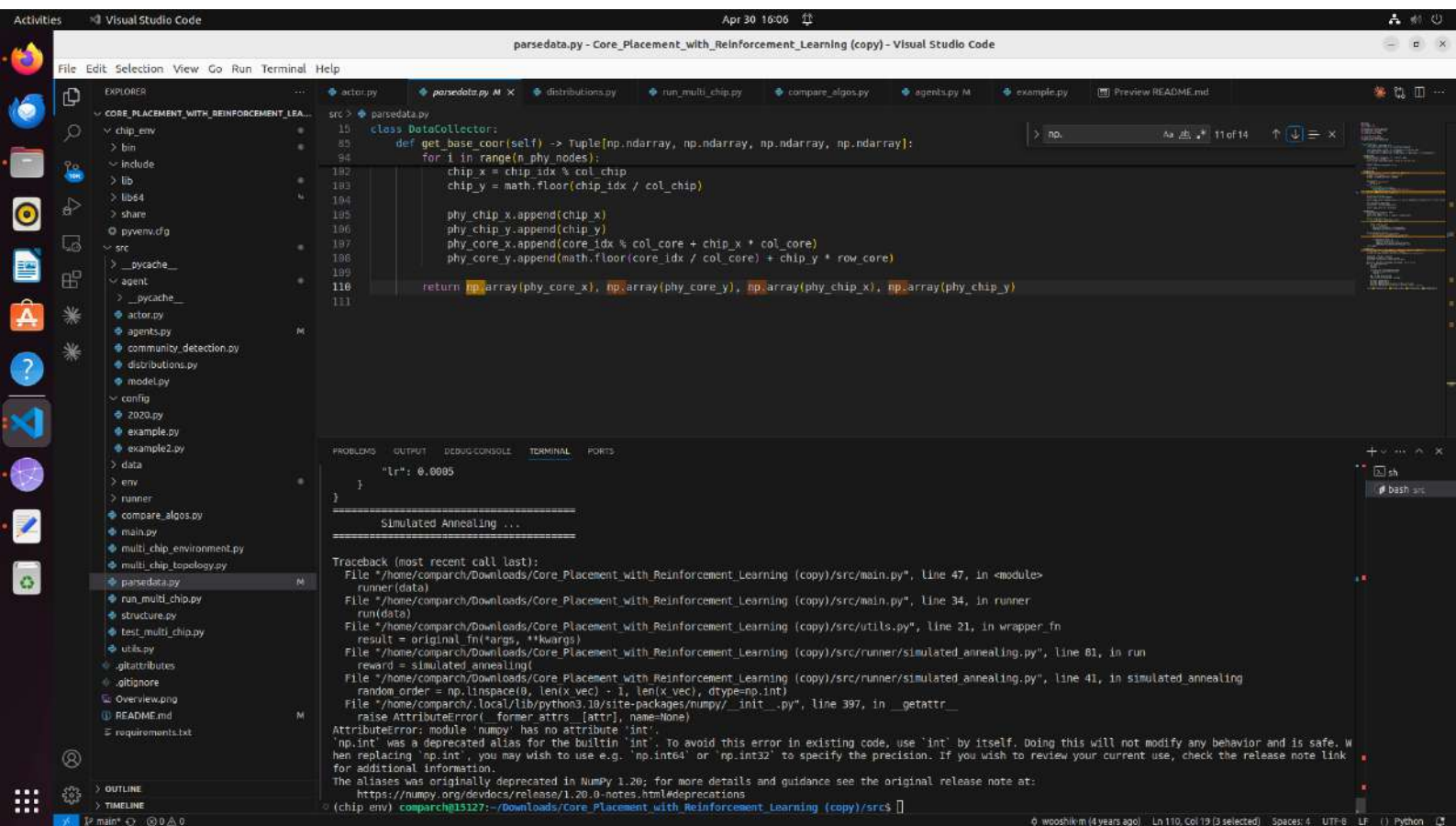
Below the epoch output, the output shows the comparison results for Simulated Annealing (SA) and PPO (Reinforcement Learning). The output is as follows:

```
=====
COMPARISON RESULTS
=====
Simulated Annealing (SA):
Final Total Cost : 65325.50
On-chip Cost      : 6276.50
Off-chip Cost     : 59049.00
Time taken        : 1.15 seconds

PPO (Reinforcement Learning):
Final Total Cost : 70461.50
On-chip Cost      : 5569.00
Off-chip Cost     : 64901.50
Time taken        : 430.39 seconds

Winner: SA beat PPO by 5136.00 cost difference!
```

The bottom status bar shows the file path: `(chip env) comparch@15127:~/Downloads/Core Placement with Reinforcement Learning (copy)/src`.



```
65 while epoch <= 100 and best_reward > 768 :
66     self.reset_buffer()
67     for _ in range(self.batch_size):
68         # reward sampling
69         with torch.no_grad():
70             action, log_prob = self.policy()
71             reward, current_grid = self.env.step(action)
72
73         # store episodes
74         self.save_epiosde(action, log_prob, reward)
75         if(reward<best_reward):
76             best_grid=current_grid
77         best_reward = min(-reward, -best_reward)
78
```

```
40 63 51 43 3 34 9 43 6 47 18 7 17 26 28 32]
(chip_env) comparch@15127:~/Downloads/Core_Placement_with_Reinforcement_Learning (copy)/src$ python3 run_multi_chip.py --algo rl --iters 100
System : MultiChipTopology(2x2 chips, 4x4 cores/chip, total=64 cores)
Tasks : 64
Algo : rl
-----
[RL] Could not import agent from src/agent.py.
      Make sure src/ is on PYTHONPATH and dependencies are installed.
      Falling back to random search.

Final placement cost : 68150.5000
On-chip comm cost : 5900.0000
Off-chip comm cost : 62250.5000
Chip placement : [56 5 40 60 15 57 29 24 55 38 20 25 27 45 18 34 35 1 11 6 23 48 61 16
17 13 59 37 12 30 19 41 4 36 28 63 51 0 3 22 43 62 32 9 31 10 49 26
14 54 21 7 33 44 50 8 46 42 52 58 53 2 39 47]
(chip_env) comparch@15127:~/Downloads/Core_Placement_with_Reinforcement_Learning (copy)/src$
```

